STUDENT INTERNSHIP PROGRAM

REPORT ON

"Cisco Networking Academy Virtual Internship Program"

Internship Provider

Cisco Networking Academy (Cisco Systems, Inc.)

Submitted by:

Shridhar Ambore

PRN: 202401040197

Exam Seat No.: 202401040197

Class: TY B.Tech, Computer Engineering | Division / Roll No: 114

Internship Duration: 120 Hours

Department of Computer Engineering

MIT Academy of Engineering, Alandi(D)

Pune Maharashtra - 412105

Academic Year 2026-27

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

CERTIFICATE

This is to certify that the **"Student Internship Program (SIP)"** report submitted by **Shridhar Ambore**, PRN- **202401040197** is work done by him and is submitted during 2026-2027 academic year.

Internship Guide	**Internship Coordinator**
(Mrs. Sharmila B. Kharat)	(Mrs. Savita Mane)

HOD
Department of Computer Engineering
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Department of Computer Engineering
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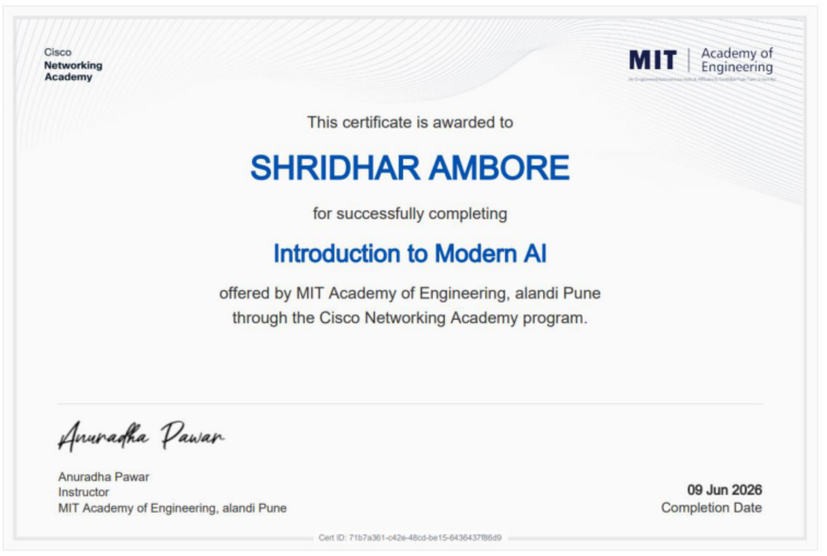
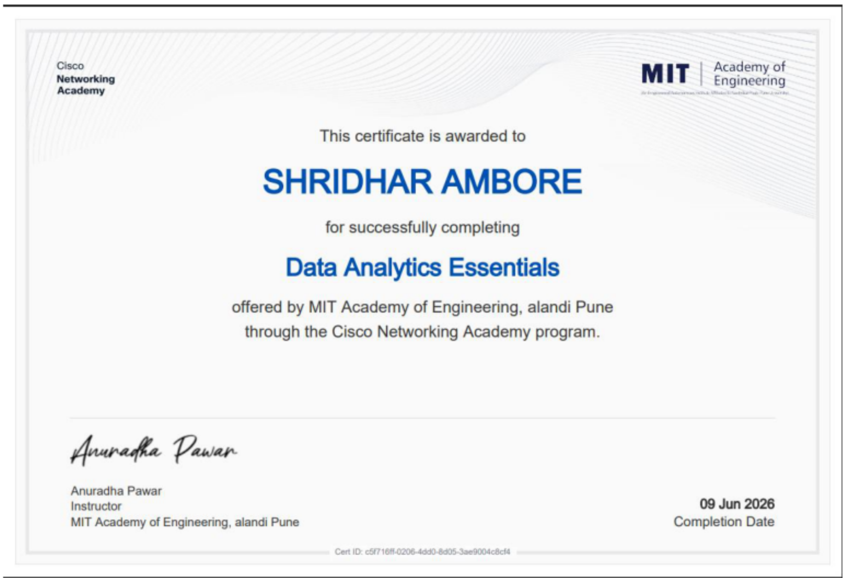
Course Completion Certificates

The internship included five Cisco Networking Academy / JS Institute courses. The completion certificates are included here as evidence of the training completed during the internship. The first three certificates were completed on 09 June 2026 and the two JavaScript certificates were completed on 26 June 2026.

Internship Certificate





Cisco Networking Academy

MIT | Academy of Engineering

This certificate is awarded to

SHRIDHAR AMBORE

for successfully completing

Data Analytics Essentials

offered by MIT Academy of Engineering, alandi Pune
through the Cisco Networking Academy program.

Anuradha Pawar

Anuradha Pawar
Instructor
MIT Academy of Engineering, alandi Pune

09 Jun 2026
Completion Date

Cert ID: c5f716ff-0206-4dd0-8d05-3ae9004c8cf4

Cisco Networking Academy

JS INSTITUTE
Open Education & Development Group

This certificate is awarded to

SHRIDHAR AMBORE

for successfully completing

JavaScript Essentials 1

offered by Networking Academy
through the Cisco Networking Academy program.

Lynn Bloomer

Lynn Bloomer
Director
Cisco Networking Academy

26 Jun 2026
Completion Date

Cert ID: 8ed850d1-d1ec-4c2c-bf40-931d760f40cd

Cisco Networking Academy | JS INSTITUTE Open Education & Development Group

This certificate is awarded to

SHRIDHAR AMBORE

for successfully completing

JavaScript Essentials 2

offered by Networking Academy
through the Cisco Networking Academy program.

Lynn Bloomer
Lynn Bloomer
Director
Cisco Networking Academy

26 Jun 2026
Completion Date

Cert ID: 84bf6b0f-8059-404f-9b99-8f32b840907b

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to MIT Academy of Engineering, Alandi (D) for providing me with the opportunity to undertake the Student Internship Program (SIP). The internship provided an important opportunity to connect academic concepts with current industry-oriented learning in Artificial Intelligence and Data Analytics.

I am thankful to Cisco Networking Academy for providing the learning platform, structured courses, practice activities, assessments, and supporting resources used during the Cisco Virtual Internship Program 2026. The three courses considered in this report introduced modern AI concepts, practical AI-assisted customer-review analysis, and the foundations of data analytics.

I express my heartfelt gratitude to my Internship Guide, Mrs. Sharmila B. Kharat, for her guidance, encouragement, and suggestions during the internship and preparation of this report. I also thank Mrs. Savita Mane and Dr. Supriya Sabale for their coordination and support.

I extend my sincere thanks to Dr. Pramod Ganjewar, Head of the Department of Computer Engineering, for providing the academic environment and encouragement required to complete the internship successfully. Finally, I thank all faculty members and everyone who contributed directly or indirectly to this work.

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Abstract

The Cisco Virtual Internship Program 2026 provided a structured introduction to modern Artificial Intelligence and Data Analytics through Cisco Networking Academy learning resources. This report documents the knowledge, practical exercises, and engineering understanding developed during the internship. The learning was organized around three courses: *Introduction to Modern AI*, *Apply AI: Analyze Customer Reviews*, and *Data Analytics Essentials*.

The first course established foundational AI literacy through concepts of Artificial Intelligence, Machine Learning, Generative AI, computer vision, machine translation, chatbots, and effective prompting. The second course focused on applying AI to tabular customer-review data, with emphasis on thematic analysis, spreadsheet-based work, AI tool selection, and prompt-driven extraction of useful information. The third course developed the fundamentals of data analytics through project organization, data gathering, cleaning, Excel transformation, basic statistics, relational databases, SQL, structured queries, Tableau visualization, ethics, and career development.

The report also documents the applied customer-review analytics project associated with the internship context. The supplied system architecture demonstrates a workflow in which CSV/Excel reviews are uploaded, cleaned with Pandas, stored in PostgreSQL, processed through background AI tasks, analyzed using Gemini/OpenRouter services, and presented through a Chart.js dashboard. The project section is presented as an application of the course ideas rather than as a replacement for the official course curriculum.

Overall, the internship strengthened understanding of AI-assisted workflows, data preparation, analytical reasoning, structured querying, visualization, responsible use of AI, and the importance of selecting an appropriate tool for a given task.

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Introduction

1.1 Internship Overview

The rapid growth of Artificial Intelligence and data-driven decision making has changed the skills expected from computer engineering students. AI is no longer restricted to research laboratories; AI-enabled applications are now present in search, translation, photography, productivity software, customer support, education, and business analytics. At the same time, organizations increasingly depend on data analysts who can gather, prepare, query, visualize, and communicate information.

The Cisco Virtual Internship Program 2026 was undertaken as a structured learning experience in these areas. The report focuses on three Cisco Networking Academy courses: *Introduction to Modern AI*, *Apply AI: Analyze Customer Reviews*, and *Data Analytics Essentials*. The first two courses introduce practical AI literacy and AI-assisted analysis, while Data Analytics Essentials develops a broader analytical workflow using Excel, SQL, Tableau, and basic statistical methods.

Table 1.1: Internship Overview

Parameter	**Details**
Program	Cisco Virtual Internship Program 2026
Provider	Cisco Networking Academy
Student	Shridhar Ambore
PRN	202401040197
Duration	120 Hours; course certificates completed in June 2026
Main Areas	Modern AI, applied AI, customer-review analysis, data analytics

1.2 Objectives of the Internship

The internship objectives can be grouped into conceptual, practical, and professional objectives. Conceptually, the aim was to understand basic AI terminology and the relationship between AI, machine learning, generative AI, computer vision, machine translation, and chatbots. Practically, the aim was to use AI tools appropriately, write useful prompts, analyze tabular text data, and understand a complete data analytics workflow. Professionally, the objective was to develop analytical thinking, responsible AI awareness, and confidence in using industry-relevant tools.

1. Understand the fundamental terminology used in modern AI.

2. Recognize how machine learning and generative AI fit into the wider AI field.
3. Explore practical applications of computer vision and machine translation.
4. Develop effective prompts for common chatbot tasks.
5. Apply AI-assisted thematic analysis to customer-review data.
6. Understand data gathering, cleaning, transformation, analysis, and visualization.
7. Learn the role of Excel, SQL, relational databases, and Tableau in analytics.
8. Recognize ethical issues, bias, privacy, and limitations of AI and analytics.

1.3 Scope of the Report

The report is intentionally limited to the learning themes of the three internship courses and the associated customer-review analytics application. It does not present advanced model training, production-scale machine learning engineering, or unrelated networking topics as internship outcomes. This distinction is important because the purpose of the report is to accurately document what was learned rather than to turn a beginner-level course into an advanced AI engineering project.

1.4 Report Organization

The report begins with the organization and learning environment, followed by separate chapters for the three courses. Later chapters discuss practical AI usage, data analytics workflow, the customer-review analytics project, challenges, learning outcomes, and future scope. The conclusion summarizes the overall value of the internship.

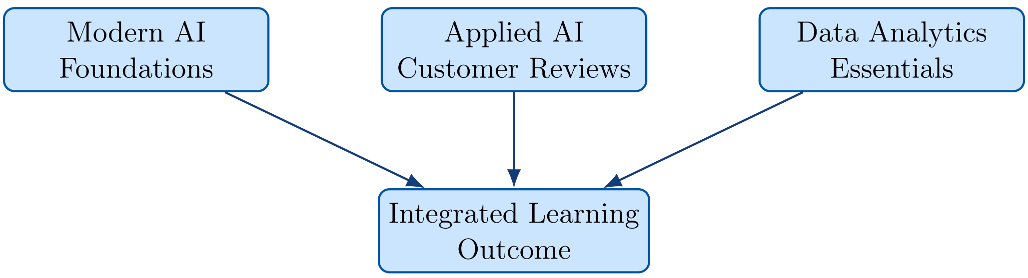

Figure 1.1: Overall Learning Structure of the Internship

Chapter 2

Organization Overview

2.1 About Cisco

Cisco is a global technology company known for networking, security, collaboration, observability, and related software and services. Its history is strongly associated with connecting different networks, but its current technology portfolio extends into cloud, security, automation, analytics, and AI-enabled capabilities. This broader context is relevant to the internship because modern computing systems increasingly combine networking, software, data, and intelligent services.

2.2 Cisco Networking Academy

Cisco Networking Academy is an educational initiative that provides structured technology learning through online and instructor-supported courses. Its course portfolio covers areas such as networking, cybersecurity, programming, IT, AI, and data science. The learning approach combines explanations with practice opportunities and assessments. For a student, this structure is useful because it breaks a large technical field into manageable competencies.

2.3 Learning Philosophy

A major strength of the Networking Academy model is its emphasis on practical understanding. Instead of learning terminology in isolation, learners are exposed to scenarios where a tool or concept solves a particular problem. In the AI courses, this means experimenting with prompts, understanding AI-enabled applications, and analyzing text. In Data Analytics Essentials, the same principle appears through spreadsheets, SQL queries, visualization, and project-based analysis.

2.4 Relevance to Computer Engineering

Computer engineering combines programming, systems, databases, algorithms, and practical problem solving. The internship connects these foundations with modern AI-assisted workflows. For example, a student who understands tables and databases can better understand analytics; a student who understands programming can reason about automation; and a student who understands algorithms can better interpret what an AI system is doing. The internship therefore acts as a bridge between academic computing and modern digital tools.

Table 2.1: Connection Between Internship Areas and Computer Engineering

Internship Area	**Engineering Connection**	**Practical Value**
Modern AI	Algorithms, software tools	AI literacy and tool selection
Applied AI	NLP concepts, structured data	Automated review analysis
Data Analytics	Databases, statistics, visualization	Evidence-based decisions
Ethics	Software responsibility	Safer technology use

2.5 Internship Course Completion Record

The internship was completed through five certification courses. The certificates included in this report confirm the completion of the courses listed below.

Table 2.2: Courses Completed During the Internship

Course	**Status**	**Certificate Date**
Introduction to Modern AI	Completed	09 June 2026
Data Analytics Essentials	Completed	09 June 2026
Apply AI: Analyze Customer Reviews	Completed	09 June 2026
JavaScript Essentials 1	Completed	26 June 2026
JavaScript Essentials 2	Completed	26 June 2026

2.6 Internship Learning Environment

The virtual format required independent progress through learning material, practice tasks, and assessments. Such an environment develops a different skill from classroom instruction: the learner must identify what is understood, revisit unclear concepts, and use documentation or examples when necessary. This is particularly valuable in AI, where tools and terminology evolve rapidly.

Chapter 3

Course I – Introduction to Modern AI

3.1 Course Purpose

Introduction to Modern AI is a beginner-friendly course designed to build AI literacy and practical confidence. Cisco describes the course as requiring no coding and focusing on how learners can use AI in daily life. The official course description highlights an overview of AI, computer vision, machine translation, chatbots, and effective prompting. It also includes hands-on chatbot exercises and an open-ended project. These themes form the basis of this chapter. [1]

3.2 Artificial Intelligence and Machine Learning

Artificial Intelligence is a broad field concerned with systems that perform tasks requiring capabilities commonly associated with human intelligence. Machine Learning is one approach within AI in which models learn patterns from data. The distinction matters because an AI application may use machine learning internally without requiring the end user to understand or implement the underlying algorithm.

A useful mental model is to view a machine-learning system as a pattern learner. Data is provided to a learning algorithm, the algorithm produces a model, and the model is then used for inference. The quality of the result depends on the data, the task definition, the model, and the way the output is interpreted.

Table 3.1: Basic AI Vocabulary

Term	**Meaning**
Artificial Intelligence	Broad field of systems performing tasks associated with intelligent behavior.
Machine Learning	Methods that learn patterns from examples or data.
Model	Learned computational representation used to produce outputs.
Generative AI	AI that produces new content such as text or images.
Prompt	Instruction or input provided to a generative AI system.
Chatbot	Conversational interface that uses language models or related AI systems.

3.3 Computer Vision

Computer vision allows software to interpret information contained in images or video. In a practical AI literacy course, the important learning outcome is not to train a vision model from scratch but to understand what vision-enabled applications can do and what limitations they have. Examples include object detection, identifying visual similarity, image segmentation, and optical character recognition.

For example, an OCR application can transform text visible in an image into editable text. A visual-search application can compare an input photograph with known visual patterns. Such tools can support education, accessibility, document processing, and everyday productivity. However, users should not assume that every detected object or extracted word is correct. Lighting, image quality, unusual viewpoints, and ambiguous content can affect results.

3.4 Machine Translation

Machine translation converts text from one language to another using computational models. Modern translation systems use context rather than simply replacing individual words. Translation quality can still be affected by word order, idioms, cultural expressions, ambiguity, domain-specific terminology, and missing context.

The course makes translation a practical AI literacy topic because it demonstrates both the power and limitations of AI. A translated sentence may be grammatically correct but still fail to preserve a cultural meaning. Therefore, human review remains important when accuracy has significant consequences.

3.5 Chatbots and Generative AI

Chatbots provide a conversational interface to AI systems. Modern generative AI systems can produce text, summarize documents, transform content, answer questions, and assist with planning. The course introduces learners to how chatbots work at a conceptual level and how better prompts can produce more useful results.

Generative AI differs from conventional search in an important way. A search system primarily retrieves existing information, while a generative system constructs an output based on learned patterns and the current input. This makes generative AI flexible, but it also means that an answer can sound convincing without necessarily being correct. Users must therefore evaluate important outputs rather than treating fluent language as proof of accuracy.

3.6 Prompting Principles

Prompting is the process of giving instructions that guide an AI system toward a desired output. Effective prompting usually benefits from specifying the task, providing context, identifying constraints, and describing the expected format. For example, a request to "summarize this document" is less constrained than a request to "summarize the document in five bullet points for a first-year engineering student and identify two action items."

The course explores practical chatbot use through multiple task types. These include expanding an idea, synthesizing information, transforming text, evaluating content, carrying out a conversation, taking action through a workflow, and working with images or text as input. This approach demonstrates that prompting is not merely about asking questions; it is about defining a useful interaction.

Table 3.2: Prompting Techniques and Their Uses

Technique	**Purpose**	**Example Use**
Clear task	Defines the required action	Summarize a report
Context	Reduces ambiguity	Explain for a beginner
Constraints	Controls the result	Limit to five bullets
Format	Makes output reusable	Return a table
Examples	Demonstrates desired behavior	Few-shot classification
Iteration	Improves weak outputs	Refine an initial prompt

3.7 Understanding Model Limitations

A central lesson of AI literacy is that AI tools are useful but not infallible. Hallucination refers to generated information that is unsupported or incorrect. Models can also have limitations involving context length, current information, ambiguity, bias, and image interpretation. The correct response is not to reject AI entirely but to use it according to the risk of the task.

For low-risk tasks such as brainstorming, rewriting, or generating alternative explanations, an imperfect answer can often be corrected easily. For high-impact decisions, factual verification and human judgment are necessary. This risk-based approach is an important professional habit.

3.8 Open-Ended Project and Learning Outcome

The course includes an open-ended project that allows learners to demonstrate practical AI usage. The important outcome is the ability to select a suitable AI-enabled tool, formulate a task, communicate effectively with the tool, and evaluate the result. This is a transferable skill because the exact AI application may change while the reasoning process remains useful.

3.9 Course I Summary

The first course developed foundational AI literacy rather than advanced model development. The most important outcomes were understanding basic AI concepts, recognizing computer vision and machine translation applications, learning how chatbots can assist with everyday tasks, practicing effective prompts, and understanding the limitations of AI-generated outputs. These foundations support the more specific applied analysis studied in the next course.

Chapter 4

JavaScript Essentials I & II

4.1 Course Overview

JavaScript Essentials I and II were completed through the JS Institute in collaboration with Cisco Networking Academy. These courses were important for the internship because the final project is not only an AI application; it also needs a browser-based interface that can communicate with the backend. Part I focused on language fundamentals, while Part II moved further into asynchronous programming, APIs, and reusable code.

4.2 JavaScript Essentials I

The first course established the basic programming model of JavaScript. I worked with variables, data types, operators, conditional statements, loops, functions, arrays, and objects. The course also introduced modern JavaScript syntax, which made the code easier to organize.

A useful part of the course was learning how functions and objects can be combined to keep application logic separate from the user interface. Arrays were especially important because browser applications frequently receive lists of records from an API.

4.3 Important Topics from JavaScript Essentials I

Table 4.1: JavaScript Essentials I – Major Topics

Topic	**What I Learned**
Variables and data types	Store and work with different kinds of values.
Operators and conditions	Build decisions and logical expressions.
Loops	Repeat operations over data.
Functions	Break application logic into reusable units.
Arrays	Store and process collections of values.
Objects	Represent structured application data.
Modern ES6+ syntax	Use let, const, arrow functions and other modern syntax.
DOM basics	Read and update elements displayed in the browser.
Events	Respond to actions such as clicks and form submission.

4.4 JavaScript Essentials II

The second course moved from basic syntax toward the way JavaScript is used in real web applications. The most useful topics for my project were asynchronous programming, promises, async/await, REST APIs, and organizing code into reusable modules.

The main difference I noticed between a small JavaScript program and a web application is that the browser often has to wait for something outside the page. A request may be sent to a server and the result may arrive later. JavaScript therefore needs a way to continue running without freezing the interface.

4.5 Asynchronous Programming

Callbacks, promises, and async/await were covered as different ways of handling asynchronous operations. In the project, the same idea appears when the browser uploads a file, starts analysis, and then checks the server for progress.

```
async function checkStatus(taskId) {
    const response = await fetch(‘/api/status/${taskId}‘);
    const status = await response.json();
    updateProgress(status);
}
```

This pattern is representative of the front-end behavior of the dashboard. The page sends a request, waits for the response, and updates only the part of the interface that has changed.

4.6 Working with REST APIs

The course introduced communication with a server through HTTP requests. This became directly useful in the dashboard because the frontend and Flask backend are separate layers. The browser sends requests to endpoints such as `/api/upload`, `/api/analyze`, `/api/status`, and `/api/result`.

The JavaScript layer collects user input, sends the request, interprets the JSON response, and changes the displayed state. The frontend therefore does not need to know how the Python backend performs cleaning or AI processing.

4.7 DOM Manipulation and Events

The DOM represents the structure of the web page. JavaScript can find elements, change their contents, show or hide sections, and respond to user actions. In the project this is useful for showing upload information, displaying analysis progress, enabling result views, and updating dashboard components after the backend returns data.

4.8 Modules and Reusable Code

JavaScript Essentials II also emphasized organizing code into reusable pieces instead of placing all logic in one large script. This approach is useful for a dashboard because file upload, status polling, result rendering, and export actions are different responsibilities.

4.9 Practical Application in the Project

The JavaScript courses contributed mainly to the user-facing side of the AI Customer Review Analytics Dashboard. The user can select a CSV or Excel file, preview the cleaned data, start analysis, and observe the progress while the backend works in the background. Once the analysis is complete, the page receives the results and passes the structured data to Chart.js for visualization.

This was a useful reminder that an AI application is not only the model call. A usable system needs an interface, API communication, error handling, asynchronous behavior, and a way to present the result.

4.10 Learning Outcome

By completing JavaScript Essentials I and II, I became more comfortable writing modern JavaScript and understanding how a browser communicates with a backend service. The most useful outcomes for the internship project were DOM interaction, event handling, asynchronous requests, promises, async/await, JSON data handling, and REST API communication.

Chapter 5

Course II – Apply AI: Analyze Customer Reviews

5.1 Course Purpose

The second course moves from general AI literacy to a specific analytical scenario: using AI to analyze customer reviews. Cisco's verified learning outcome describes thematic analysis of tables of text with assistance from AI and spreadsheet applications, along with choosing the right AI or non-AI tool for each task. The associated skills include AI tool selection, LLM prompting, spreadsheet formulas, tabular data, and thematic analysis. [2]

5.2 Customer Reviews as Data

Customer reviews are an example of semi-structured or unstructured textual information that can contain valuable business signals. A review may discuss product quality, price, delivery, customer service, usability, or several topics at the same time. Manually reading hundreds or thousands of reviews is time-consuming. AI assistance can help identify recurring themes and organize information for further analysis.

However, the goal is not simply to ask an AI model for a general summary. Useful analysis requires a clear question and an output that can be compared across many rows. This is where structured prompts and spreadsheets become important.

5.3 Thematic Analysis

Thematic analysis identifies recurring topics or themes in textual data. Consider three hypothetical reviews: one customer may complain about delayed delivery, another may praise product quality, and another may report a poor support experience. These statements can be grouped under themes such as Delivery, Product Quality, and Customer Service.

The value of thematic analysis is that it converts many individual comments into patterns that a business can investigate. A theme does not automatically prove a root cause; it is evidence that a particular issue or topic appears repeatedly and deserves attention.

5.4 Choosing AI or Non-AI Tools

A particularly useful course concept is tool selection. Not every problem requires generative AI. Spreadsheet formulas are often better for deterministic calculations such as

Table 5.1: Illustrative Customer-Review Thematic Analysis

Review Example	Theme	Possible Business Question
Delivery arrived two days late.	Delivery	Is a logistics region causing delays?
The product feels durable and well made.	Product Quality	Which product features are most appreciated?
Support took too long to answer.	Customer Service	Where can response time be improved?
The price is high compared with alternatives.	Pricing	Is pricing affecting customer satisfaction?

totals, counts, averages, and conditional logic. SQL is appropriate when information is stored in relational tables and needs structured retrieval. A chatbot is useful when the task involves language, summarization, classification of text, or transformation of content.

This principle can be summarized as "use AI where it adds value." Using a language model for a simple arithmetic operation may introduce unnecessary uncertainty. Conversely, asking a spreadsheet formula to understand the meaning of a paragraph is inappropriate. Effective analysts combine tools instead of assuming that one tool is best for every task.

5.5 Prompt Design for Reviews

A review-analysis prompt should define the role, task, review text, categories if required, and desired output. A useful structure is: "Analyze the following review. Identify the main theme, classify the sentiment if requested, and return the result in a fixed table with one row per review." For larger datasets, examples can be provided so that the model understands the expected categories.

Prompt quality affects consistency. If categories are not defined, the model may use different labels such as "shipping", "delivery", and "logistics" for similar ideas. A controlled vocabulary can reduce this variation. Similarly, specifying the output format makes the result easier to move into a spreadsheet or database.

5.6 Spreadsheet-Assisted Analysis

The course combines AI with spreadsheet work because tabular data requires organization. A spreadsheet can hold the original review, extracted theme, sentiment or category where applicable, and other structured attributes. Formulas can count occurrences, calculate percentages, and support filtering. This combination gives the learner both qualitative and quantitative views of the review dataset.

For example, once themes are assigned, a count of each theme can reveal which topics occur most frequently. A pivot table can summarize the number of reviews per theme. A chart can then communicate the distribution to a decision maker. AI assists with interpretation of text; spreadsheet tools assist with deterministic organization and aggregation.

5.7 Quality Checking

AI-generated analysis should be checked. Errors can occur because a review is ambiguous, sarcastic, too short, or contains multiple issues. A quality-control process can include reviewing a sample of outputs, checking category consistency, looking for unexpected labels, and comparing AI results with human judgment.

A practical workflow is therefore iterative: define the categories, test the prompt on a small sample, inspect the outputs, revise the instructions, and then process the larger dataset. This is more reliable than immediately applying an untested prompt to every row.

5.8 Course II Summary

The second course demonstrated a concrete use of AI in business analysis. Its major lessons were thematic analysis, structured prompting, spreadsheet-assisted organization, tool selection, and output checking. These concepts provide a direct bridge to the customer-review analytics project documented later in this report.

Chapter 6

Course III – Data Analytics Essentials

6.1 Course Purpose

Data Analytics Essentials develops the fundamental workflow and tools used by a data analyst. Cisco's course description emphasizes transforming, organizing, and visualizing data with Excel, querying relational databases with SQL, and improving data presentations with Tableau. The current course structure includes data analytics projects, data gathering and investigation, data preparation and cleaning, Excel transformation, statistics, relational databases and SQL, structured queries, Tableau, ethics and bias, and next steps. [3]

6.2 Module 1: Data Analytics Projects

A data analytics project begins with a question or business requirement. The analyst must understand what decision the analysis is expected to support. A project is therefore more than making charts. It involves defining the problem, identifying relevant data, preparing the data, analyzing it, communicating the results, and reflecting on limitations.

A clear project question helps prevent irrelevant analysis. For example, "What are the most common customer complaints?" is more useful than simply asking for "interesting patterns." The first question guides data selection and defines the expected output.

6.3 Module 2: Data Gathering and Investigation

Data may be obtained from files, databases, applications, surveys, sensors, or other sources. Before analysis, the analyst should investigate the structure and quality of the available information. Important questions include: What does each column represent? Which values are missing? Are units consistent? Are there duplicate records? What time period does the data cover?

Investigation is an important analytical habit because incorrect assumptions about data can produce misleading conclusions. A column containing a numeric value may represent a code rather than a measurable quantity. A date may be stored as text. A blank value may mean "not applicable" rather than "unknown." Understanding these details comes before transformation.

6.4 Module 3: Preparing and Cleaning Data

Data cleaning prepares information for analysis. Common operations include removing duplicate records, addressing missing values, correcting inconsistent formats, identifying outliers, and validating data types. Cleaning should be documented because changes to the dataset can affect the final results.

Missing data requires judgment. If only a few values are missing and the field is not critical, a suitable replacement may be possible. If a large part of a column is missing, the analyst may need to reconsider whether the field is useful. There is no universal rule; the decision depends on the meaning of the variable and the analytical objective.

6.5 Module 4: Transforming Data with Excel

Excel is a practical tool for organizing and transforming tabular data. Important operations include sorting, filtering, formulas, conditional calculations, lookup functions, and pivot tables. These features allow an analyst to move from raw rows toward a dataset suitable for analysis.

Pivot tables are especially useful because they summarize data without requiring complex programming. For example, a movie dataset can be grouped by genre and year, or a sales dataset can be summarized by product category. Formulas can calculate derived values, while lookup functions can connect related information across tables.

Table 6.1: Examples of Spreadsheet Operations

Operation	**Purpose**	**Example**
Sort	Arrange records	Highest rating first
Filter	Select relevant rows	Movies released after 2020
Formula	Calculate a value	Profit = Revenue - Cost
Lookup	Retrieve related data	Match an identifier to a category
Pivot Table	Summarize records	Count titles by genre
Chart	Communicate patterns	Rating distribution

6.6 Module 5: Statistics for Analysis

Statistics provides methods for describing and interpreting data. Measures such as mean, median, minimum, maximum, range, and frequency help summarize a dataset. The mean can be strongly affected by extreme values, whereas the median can provide a more robust description when a distribution is skewed.

An analyst should distinguish between a pattern and a conclusion. A correlation between two variables does not automatically mean that one causes the other. Similarly, a single outlier should not be removed merely because it looks unusual; it should be investigated to determine whether it represents an error or a legitimate observation.

6.7 Module 6: Relational Databases and SQL

Relational databases organize information into tables consisting of rows and columns. Relationships between tables are established using keys. SQL provides a standardized way to retrieve and manipulate relational data.

A simple query may select specific columns and rows:

```
SELECT title, rating
FROM movies
WHERE rating >= 8;
```

Grouping allows aggregate analysis:

```
SELECT genre, COUNT(*) AS movie_count
FROM movies
GROUP BY genre;
```

These examples illustrate why SQL is important to data analytics. Rather than exporting an entire database and manually inspecting it, an analyst can request exactly the information needed for a question.

6.8 Module 7: Structured Queries and Multiple Tables

Real datasets often contain related information in multiple tables. SQL JOIN operations combine related records using common fields. For example, a customer table can be connected to an orders table through a customer identifier. Structured queries can then filter, group, sort, and aggregate information across the combined dataset.

Table 6.2: Common SQL Concepts in Data Analytics

Concept	**Purpose**	**Typical Use**
SELECT	Choose columns	Retrieve required fields
WHERE	Filter rows	Apply conditions
ORDER BY	Sort results	Highest or lowest values
GROUP BY	Create groups	Aggregate by category
JOIN	Combine tables	Connect related records
COUNT / AVG / SUM	Aggregate values	Counts, averages, totals

6.9 Module 8: Tableau and Data Visualization

Visualization converts analytical results into a form that people can interpret quickly. Tableau provides tools for creating charts, interactive views, and dashboards. A good visualization should answer a question rather than merely decorate a report.

Chart selection depends on the analytical purpose. Bar charts are useful for comparing categories. Line charts are useful for trends over time. Scatter plots can help explore relationships between numeric variables. A dashboard combines multiple views so that users can inspect a dataset from different perspectives.

6.10 Module 9: Ethics and Bias in Data

Data analysis can be affected by sampling bias, incomplete data, measurement errors, and inappropriate assumptions. An analyst should consider who is represented in the data, who is missing, and whether the collection method introduces systematic distortion. Privacy is also important when datasets contain personal information.

Ethical analytics means communicating limitations honestly. A chart should not imply a level of certainty that the underlying data cannot support. Similarly, a model or analysis should not be used to make sensitive decisions without considering fairness and potential harm.

6.11 Module 10: Next Steps

The final stage of the course encourages continued learning and portfolio development. Data analytics is a broad field, and foundational knowledge of Excel, SQL, Tableau, statistics, and data preparation can support further study in programming, data science, business intelligence, and AI.

6.12 Course III Summary

Data Analytics Essentials provides the strongest connection between raw information and decision making. Its central lesson is a disciplined workflow: define the question, gather data, investigate it, clean and transform it, analyze it, visualize the findings, consider ethical implications, and communicate the result.

Chapter 7

Customer Review Analytics Project

7.1 Project Overview

The customer-review analytics project represents the applied portion of the internship learning. Its purpose is to transform uploaded review data into organized analytical results and dashboard-level insights. The project combines the ideas from the Apply AI course with data preparation and visualization concepts from Data Analytics Essentials.

The project architecture supplied for this report shows a web-based system in which a user uploads a CSV or Excel dataset. The backend performs cleaning with Pandas, stores data in PostgreSQL, creates a background analysis task, sends review batches to an external LLM service such as Google Gemini or OpenRouter, stores the generated results, and exposes them to a Chart.js dashboard.

7.2 System Objectives

The project has five main objectives: accept tabular review data, prepare the data for analysis, apply AI-assisted text analysis, store results reliably, and communicate findings through an interactive dashboard. The architecture separates frontend interaction, backend processing, database storage, and external AI services.

7.3 Input Data

The input is a CSV or Excel file containing customer reviews and related fields. The system is designed to tolerate common data-quality issues by cleaning the dataset before analysis. This reflects the Data Analytics Essentials principle that analysis should not begin until the analyst understands and prepares the source data.

7.4 Data Cleaning

The workflow shown in the supplied architecture removes empty rows, removes duplicate reviews, standardizes column names, handles missing values, and prepares the dataset for analysis. Pandas is used as the data-processing component. Although Pandas is not the main tool taught in Data Analytics Essentials, its role here is consistent with the broader data-preparation concept taught in the course: raw information must be transformed into a reliable analytical form.

7.5 Database Storage

PostgreSQL is used to persist upload sessions, cleaned data, analysis tasks, and results. The architecture identifies an upload-session table containing fields such as an identifier, filename, original path, cleaned data, status, and timestamps. The analysis-task table tracks task status, number of batches, completed batches, result data, insights, and timestamps.

This separation is useful because file ingestion and AI processing are not necessarily completed at the same moment. A database provides a persistent record of the state of the workflow and makes it possible for the frontend to request progress information.

7.6 Background Processing

The system uses a background worker based on Python's ThreadPoolExecutor. After an analysis task is created, the AI processing can run without blocking the main user request. The task moves through states such as pending, processing, completed, or failed. The frontend can poll a status endpoint to display progress.

This design demonstrates an important software engineering idea: long-running work should be separated from short web requests. The user interface can remain responsive while the backend processes review batches.

7.7 LLM Analysis

The architecture shows Gemini and OpenRouter as external AI services. Reviews are divided into batches and sent for analysis. The requested output includes sentiment, category, confidence score, and business insights. The exact analytical result depends on the prompt and the review content, so output validation remains important.

The project directly applies the Apply AI course concept of thematic analysis. Instead of asking the model for an unstructured paragraph, the workflow requests structured information that can be stored and visualized. This is an example of prompt engineering serving a practical data-processing objective.

7.8 Generated Results

The project generates sentiment labels such as Positive, Neutral, and Negative, categories such as Quality, Service, and Pricing, confidence values, and higher-level business insights. These fields convert text into structured analytical attributes. Once stored, the attributes can be counted, grouped, filtered, and visualized.

7.9 Dashboard and Visualization

The final stage uses Chart.js to display metrics and trends. The dashboard can visualize sentiment distribution, ratings, category frequencies, and other analytical summaries. It also provides a way to inspect reviewed data and export reports. This reflects the visualization principle from Data Analytics Essentials: the purpose of a chart is to communicate an analytical finding to a user.

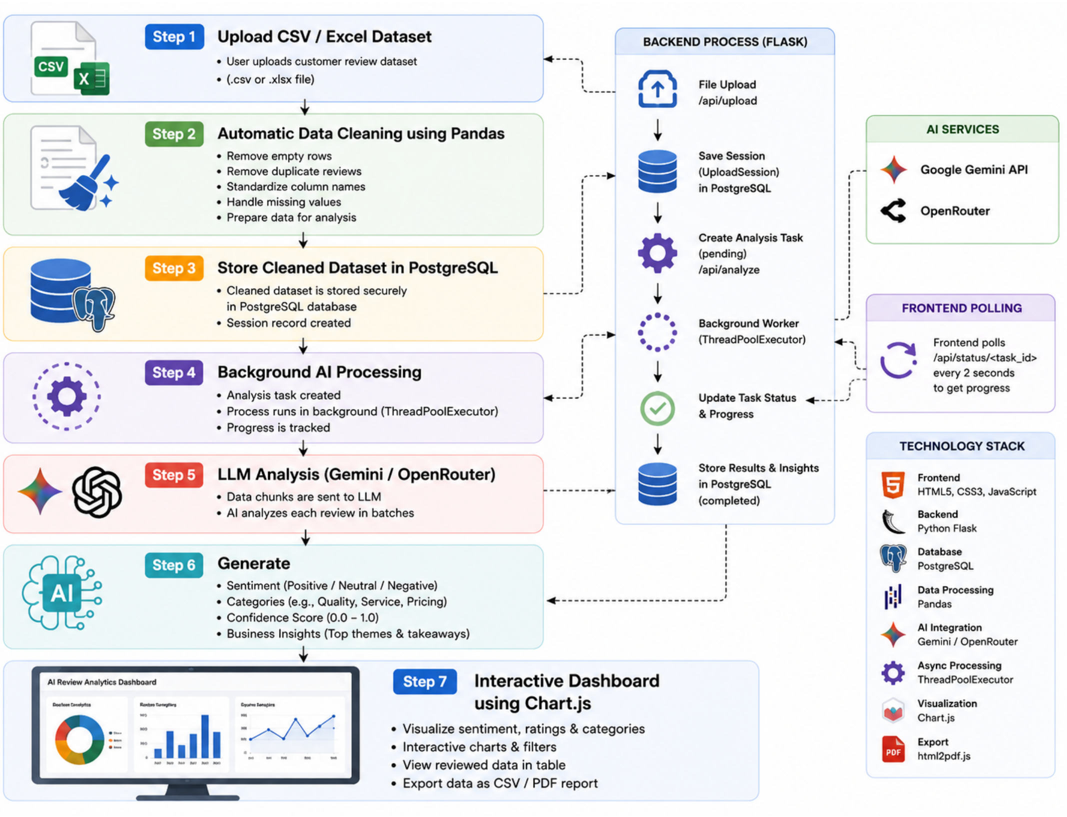

Figure 7.1: Supplied End-to-End Workflow for the AI Review Analytics Dashboard

7.10 Technology Stack

7.11 Project-to-Course Mapping

7.12 Project Learning Outcome

The project demonstrates how separate learning concepts can be combined into one workflow. Modern AI provides the understanding of AI-enabled tools and prompting. Apply AI provides the customer-review analysis scenario. Data Analytics Essentials provides the discipline of cleaning, structuring, querying, and visualizing information. The result is an integrated analytical application rather than an isolated demonstration of an AI model.

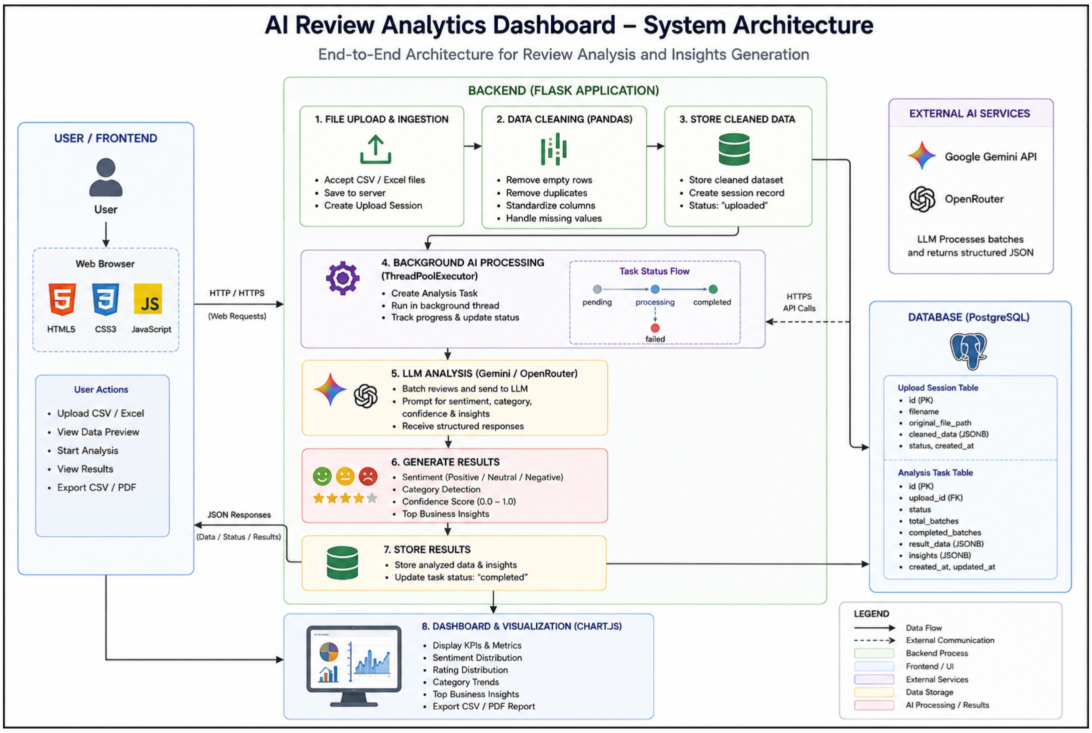

Figure 7.2: Supplied System Architecture of the AI Review Analytics Dashboard

Table 7.1: Project Technology Stack

Layer	Technology	Role
Frontend	HTML5, CSS3, JavaScript	User interface and status polling
Backend	Python Flask	HTTP endpoints and workflow control
Data Processing	Pandas	Cleaning and preparation
Database	PostgreSQL	Persistent sessions and results
AI Services	Gemini / OpenRouter	LLM-based review analysis
Concurrency	ThreadPoolExecutor	Background processing
Visualization	Chart.js	Interactive charts and dashboard

Table 7.2: Mapping of Project Components to Internship Learning

Project Component	**Learning Applied**	**Course Link**
Review text analysis	Thematic analysis and prompting	Apply AI
Data cleaning	Preparation and investigation	Data Analytics Essentials
Structured database storage	Relational data concepts	Data Analytics Essentials
AI tool use	Tool selection and prompting	Modern AI / Apply AI
Dashboard	Data visualization	Data Analytics Essentials
Validation	AI limitations and ethics	All three courses

Chapter 8

Challenges, Ethics and Limitations

8.1 Learning Challenges

One challenge in a multi-topic internship is moving between different ways of thinking. AI literacy emphasizes language, model behavior, and prompt quality. Data analytics emphasizes structure, validation, statistics, and repeatable operations. A student must learn to switch between these perspectives without confusing an AI-generated interpretation with a verified analytical result.

8.2 Prompt Variability

Generative AI systems may produce different wording for similar inputs. This makes exact reproducibility harder than a deterministic spreadsheet formula. Structured prompts, controlled categories, examples, and validation can reduce variation, but they do not eliminate the need for human review.

8.3 Data Quality

Poor-quality input produces poor-quality analysis. Missing values, duplicates, inconsistent spelling, incorrect data types, and irrelevant records can all affect conclusions. The correct response is to inspect and clean the data before drawing insights.

8.4 Hallucination and Factual Reliability

AI systems can generate statements that appear plausible but are unsupported. This is especially important when customer-review analysis is used to generate business recommendations. The system should preserve the original review so that an analyst can trace an insight back to its source.

8.5 Bias

Bias can enter through data collection, sampling, labeling, model behavior, or interpretation. If a review dataset contains feedback from only a particular group of users, conclusions may not generalize to all customers. Analysts should therefore consider coverage and representation before making broad claims.

8.6 Privacy

Customer-review datasets may contain names, contact details, order information, or other personal information. Such information should not be exposed unnecessarily to external AI services. Data minimization, access control, and removal of unnecessary personal information are appropriate safeguards.

8.7 Human Oversight

AI should support decision making rather than automatically become the decision maker in situations where errors can cause meaningful harm. Human oversight is particularly important when results are ambiguous, when a recommendation has financial or reputational consequences, or when personal information is involved.

8.8 Ethical Analytics Checklist

1. Identify the purpose of the analysis.
2. Use only the data required for that purpose.
3. Check data quality and representativeness.
4. Avoid unnecessary exposure of personal information.
5. Validate important AI-generated classifications.
6. Distinguish correlation from causation.
7. Communicate uncertainty and limitations.
8. Keep a human review step for high-impact decisions.

Chapter 9

Skills Learned and Professional Development

9.1 Technical Skills

The internship developed a set of complementary technical skills. Modern AI improved familiarity with AI vocabulary, chatbots, prompting, computer vision, and machine translation. Apply AI developed thematic analysis, AI tool selection, spreadsheet-supported review analysis, and structured prompting. Data Analytics Essentials developed spreadsheet operations, statistics, SQL, relational database concepts, visualization, dashboards, and data ethics.

Table 9.1: Skills Developed During the Internship

Skill	**Evidence in Learning**	**Application**
AI literacy	AI/ML/GenAI concepts	Selecting AI tools
Prompting	Structured instructions and iteration	Chatbots and review analysis
Thematic analysis	Customer-review course	Text-based business insights
Excel	Transformation and summaries	Tabular analysis
SQL	Structured queries and joins	Database analysis
Statistics	Descriptive analysis	Interpreting distributions
Visualization	Tableau/dashboard concepts	Communicating findings
Ethics	Bias and responsible use	Safer analysis

9.2 Analytical Thinking

A major professional outcome is a more structured approach to problems. Instead of immediately selecting a tool, the learner can begin by asking what the question is, what data is available, what type of operation is needed, and how the result will be validated. This mindset is useful across software development, data analysis, and AI-assisted applications.

9.3 Problem Solving

The internship required understanding new concepts and applying them in unfamiliar contexts. A practical problem-solving cycle is to define the issue, break it into smaller tasks, select an appropriate tool, test the result, identify errors, and refine the approach. This cycle is visible in both prompt engineering and data preparation.

9.4 Communication

Data analytics is not complete when a query or calculation is correct. Results must be communicated to people who may not know the underlying technical details. Tables, charts, concise explanations, and clearly stated limitations help convert technical work into useful information.

9.5 Independent Learning

Because the internship was virtual and self-paced, time management and independent learning were important. The student had to maintain continuity across multiple courses and connect concepts that were introduced in different contexts. This is directly relevant to a technology career, where documentation, tutorials, and new tools must frequently be learned independently.

9.6 Professional Responsibility

Responsible technology use is a professional skill. AI can increase productivity, but it also creates new responsibilities concerning verification, privacy, bias, and transparency. Data analytics similarly requires careful handling of assumptions and uncertainty. The internship therefore contributed not only technical skills but also awareness of how those skills should be used.

Chapter 10

Applications and Future Scope

10.1 Applications of Modern AI

The concepts from Introduction to Modern AI can be applied to writing assistance, translation, image understanding, learning support, brainstorming, and productivity. The key principle is that the user should understand what the tool is capable of and what kind of verification is required.

10.2 Applications of Customer-Review Analysis

AI-assisted thematic analysis can help businesses organize large volumes of customer feedback. Common themes can support product improvement, service-quality investigations, and prioritization of issues. The analysis is most useful when combined with quantitative summaries such as counts, percentages, and trends.

10.3 Applications of Data Analytics

Data analytics is used in sales, finance, education, healthcare, manufacturing, logistics, entertainment, and many other sectors. The specific tools may differ, but the workflow remains similar: understand the question, gather data, prepare it, analyze it, visualize it, and communicate the result.

10.4 Future Improvements to the Project

The customer-review analytics project can be extended while preserving the core learning objectives. Possible improvements include stronger validation of AI output, better category management, more detailed filtering, historical trend analysis, role-based access, improved export functionality, and additional dashboard views. Any such extension should continue to follow the principles of data quality, appropriate tool selection, and responsible AI.

10.5 Future Learning Roadmap

The internship provides a foundation rather than an endpoint. A logical next step is to strengthen Excel and SQL skills, learn Python for data analysis, study statistics more deeply, and then explore machine learning and natural language processing. Later, more advanced topics such as model evaluation, retrieval-augmented generation, APIs, and AI application development can be studied after the fundamentals are secure.

Table 10.1: Suggested Learning Roadmap After the Internship

Stage	**Focus**	**Outcome**
1	Excel and SQL practice	Strong analytical foundation
2	Python and data handling	Programmatic data analysis
3	Statistics	Better interpretation of results
4	Machine Learning basics	Understanding predictive models
5	NLP and LLM applications	Advanced text analytics
6	AI application development	Build reliable AI-enabled systems

Chapter 11

Overall Internship Outcomes

11.1 Achievement Summary

The internship achieved its central purpose of building practical awareness of AI and data analytics. The learner gained an understanding of how AI-enabled applications work at a conceptual level, how prompts can be designed for useful interactions, how customer-review text can be organized into themes, and how analytical data can be transformed into structured findings.

11.2 Course-Wise Outcomes

Table 11.1: Course-Wise Learning Outcomes

Course	**Outcome**
Introduction to Modern AI	AI fundamentals, computer vision, machine translation, chatbot use, prompting, limitations, and practical AI literacy.
Apply AI: Analyze Customer Reviews	Thematic analysis, AI/non-AI tool selection, prompting, spreadsheet-assisted text analysis, and structured review insights.
Data Analytics Essentials	Data projects, gathering, cleaning, Excel transformation, statistics, relational databases, SQL, Tableau, ethics, and next steps.

11.3 Academic Relevance

The internship complements academic subjects such as programming, databases, statistics, software engineering, and algorithms. SQL connects directly with database coursework. Spreadsheet and statistical analysis reinforce quantitative reasoning. AI concepts provide context for modern applications of algorithms and software. The project demonstrates how these ideas can be combined into a working analytical workflow.

11.4 Industry Relevance

The ability to work with AI tools and data is increasingly useful across software and technology roles. A graduate may not become an AI specialist immediately, but AI literacy helps in using modern development and productivity tools responsibly. Data analytics skills provide a foundation for roles involving reporting, business intelligence, data preparation, and decision support.

11.5 Reflection on Learning

The most important lesson from the internship is that technology selection should follow the problem. AI is powerful for language-oriented tasks, but spreadsheets and SQL are often better for deterministic analysis. Visualization is needed to communicate results, while human judgment is required to interpret them responsibly. This combination produces more reliable systems than relying on any single tool.

11.6 Final Outcome

The internship therefore produced a balanced foundation: AI literacy, applied text analysis, data analytics, visualization awareness, and responsible technology use. These outcomes are appropriate for a computer engineering student preparing for further study and practical work in AI-enabled software and data-driven systems.

Chapter 12

Conclusion

The Cisco Virtual Internship Program 2026 was a valuable learning experience that connected modern Artificial Intelligence with practical data analytics. The three courses provided complementary perspectives: Introduction to Modern AI established foundational AI literacy; Apply AI: Analyze Customer Reviews demonstrated a focused application of AI-assisted thematic analysis; and Data Analytics Essentials developed a disciplined approach to data gathering, preparation, transformation, SQL, visualization, and ethics.

The internship also demonstrated that effective AI usage is not simply a matter of generating an answer. A reliable workflow requires a clear question, appropriate tool selection, good data, structured instructions, validation, and responsible interpretation. The customer-review analytics project illustrates this principle by combining data cleaning, AI-assisted analysis, database storage, and dashboard visualization.

Overall, the internship strengthened the technical and analytical foundation required for further learning in AI, data analytics, and software engineering. It provides a practical starting point for deeper study of Python, statistics, machine learning, natural language processing, databases, and AI application development.

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